

Knowing Whether It Works

Task 5.1. Bedrock Evaluations, LLM-as-a-judge, RAG evaluation, golden datasets and agent assessment.

You cannot **test** a generative system, because the same input does not produce the same output. You can only **measure** it, which means statistics rather than assertions.

Traditional testing or GenAI evaluation?

TRADITIONAL TEST

Assertion

- Same input, same output, every time
- Pass or fail, with no middle
- One run is conclusive
- A failure points at a line of code

GENAI EVALUATION

Measurement

- Same input, different output
- A score on a continuum
- You need many runs and a baseline
- A drop points at a component, not a line

THE TIEBREAKER

This is why the gates in modules 2, 6 and 10 all said 'record a score, not a pass'. A quality gate on a generative system is a threshold, and thresholds need a baseline.

Offline evaluation or online monitoring?

OFFLINE

Before you ship

- A fixed golden set, run on demand
- Answers: is this release good enough?
- Controlled, repeatable, comparable
- Cannot see what real users will ask

ONLINE

After you ship

- Sampled live traffic, continuously
- Answers: is production still healthy?
- Real questions, real distribution
- Noisy, and you cannot re-run yesterday

THE TIEBREAKER

You need both. Offline catches a bad release; online catches drift in a release that was fine. Module 15's quality metrics are the online half of this module.

Five job types, and what each is for

JOB TYPE	SCORES	REACH FOR IT WHEN
Programmatic	Deterministic, reference-based	You have reference answers
LLM-as-a-judge	Nuanced quality, evaluator model	Subjective quality, high volume
Human-based	A work team's judgement	High stakes, or judge calibration
RAG: retrieve-only	The retrieval stage alone	You suspect retrieval
RAG: retrieve+generate	The whole pipeline	The product as users see it

ON THE EXAM

Three model job types and two RAG job types. Knowing retrieve-only exists is what lets you isolate a retrieval problem instead of guessing.

The automatic metrics, and what each measures

- ◆ **ROUGE** measures n-gram overlap with a reference. The summarisation metric.
- ◆ **BLEU** measures n-gram precision against a reference. The translation metric.
- ◆ **BERTScore** compares embeddings rather than words, so it credits a correct paraphrase.
- ◆ **F1 and accuracy** for classification and extraction, where an answer is right or wrong.
- ◆ All four need **reference answers**. Without them, none of these metrics can be computed at all.

ON THE EXAM

ROUGE and BLEU reward using the same words as the reference. A better answer phrased differently scores badly, which is exactly the weakness BERTScore and judges exist to address.

A publisher is choosing between three foundation models for generating article summaries. It has 800 articles, each with a human-written summary produced by its editorial team over the years.

The team wants an objective, repeatable comparison it can re-run whenever a new model appears, and wants to avoid paying editors to review output during selection. Summaries will later be reviewed by editors before publication.

Which evaluation approach fits?

- | | |
|--|--|
| <p>A A human-based evaluation job with the editorial team scoring outputs from all three models</p> | <p>B A programmatic evaluation job using the existing summaries as references, scoring with ROUGE and BERTScore</p> |
| <p>C An LLM-as-a-judge job scoring helpfulness and coherence with no reference answers</p> | <p>D Deploy each model to 33% of production traffic and compare editor edit rates</p> |

◆ PAUSE AND DECIDE

Which evaluation approach fits?

A Human evaluation by the editorial team
The most trustworthy signal and the one thing the scenario excludes on cost. It is also not repeatable on demand for each new model.

C LLM-as-a-judge without references
Would work, and it discards the strongest asset available. Judging without references is what you do when references do not exist.

B Programmatic with existing summaries as references
800 human-written summaries are a reference dataset already paid for. ROUGE suits summarisation and BERTScore credits correct paraphrase.

D Three-way production traffic split
Measures real editor behaviour, and it puts three unvalidated models in front of readers to make a selection decision.

THE SIGNAL

"Each with a human-written summary" is the sentence. Reference answers already exist, which makes programmatic evaluation both cheaper and more objective than any alternative here.

LLM-as-a-judge is how you score quality at volume when no reference answers exist. It is also a model, which means it has **biases of its own**.

Four rules for using a model as an evaluator

- ◆ **Self-preference bias is real.** A model tends to prefer output that resembles its own.
- ◆ **Use an evaluator from a different model family** than the model being evaluated. That is the primary mitigation.
- ◆ **Calibrate against a small human-rated set.** Confirm the judge agrees with people before trusting it at scale.
- ◆ **Define criteria precisely.** "Is this good?" gets you noise. "Does this cite a supporting passage?" gets you a signal.
- ◆ Judge scores are for **comparison**, not absolute truth. A 0.82 means little; 0.82 against last week's 0.79 means something.

ON THE EXAM

Cross-family judging is the detail most people miss. Evaluating a model with an evaluator from the same family systematically flatters it, and the flattery is invisible in the numbers.

A team runs a nightly LLM-as-a-judge evaluation on its assistant. The evaluator model and the model being evaluated are from the same provider and family. Scores have sat around 0.91 for four months.

A separate manual review by subject matter experts rated a sample of the same responses far lower, finding frequent unsupported claims. The team is unsure which number to believe and wants an evaluation process it can defend.

What should the team change?

- ☐ A Trust the manual review and stop running automated evaluation
- ☐ B Switch the evaluator to a model from a different family, add faithfulness as an explicit criterion, and calibrate the judge against the expert-rated sample
- ☐ C Raise the judge's scoring threshold so fewer responses pass
- ☐ D Run the evaluation more frequently to increase the sample size

◆ PAUSE AND DECIDE

What should the team change?

A Abandon automated evaluation

Throws away the only mechanism that scales. Expert review of every response is not affordable, which is why the judge existed.

C Raise the threshold

Changes where the line sits on a score that is not measuring the right thing. A biased 0.91 does not become trustworthy at 0.95.

B Cross-family evaluator, explicit faithfulness criterion, calibrate

Addresses all three faults: self-preference from same-family judging, a criterion that never checked grounding, and no calibration against humans.

D Evaluate more often

More samples of a systematically biased measurement. Frequency improves precision, not accuracy.

THE SIGNAL

Same-family judging plus a four-month flat score plus experts disagreeing is three symptoms of one cause. Option D is the classic error: treating a bias problem as a sample-size problem.

A RAG system has **two** places to fail, and one evaluation number cannot tell you which. That is why Bedrock offers a retrieve-only job.

Retrieve-only or retrieve-and-generate?

RETRIEVE-ONLY

Isolates the retriever

- Scores context relevance
- Scores context coverage against ground truth
- No generation involved at all
- Use it to prove retrieval is or is not the fault

RETRIEVE-AND-GENERATE

Scores the whole pipeline

- Generation quality and faithfulness
- Citation precision and coverage
- What the user actually experiences
- Cannot localise a failure on its own

THE TIEBREAKER

Run retrieve-only first when quality drops. If retrieval scores are healthy, the problem is downstream, and you have eliminated half the system in one job.

The RAG metrics

METRIC	ASKS
Context relevance	Were the retrieved passages relevant to the query?
Context coverage	Did retrieval find the passages ground truth says it needed?
Faithfulness	Is the answer supported by the retrieved context?
Citation precision	Are the citations given actually relevant?
Citation coverage	Are the claims made actually cited?

ON THE EXAM

Faithfulness and citation coverage catch different faults. An answer can be faithful to its context and cite none of it, failing an audit while passing a grounding check.

A legal research assistant has seen answer quality complaints rise. The team has a ground-truth dataset mapping 300 questions to the passages that should answer them.

Leadership wants to know whether to invest engineering time in the retrieval pipeline or in prompt and model work. The team has two weeks and cannot rebuild both. Nobody can currently say which half is at fault.

What should the team do first?

- ☐ A Run a retrieve-only RAG evaluation against the ground-truth passages to measure context relevance and coverage
- ☐ B Run a retrieve-and-generate evaluation to get an overall quality score
- ☐ C Re-ingest the corpus with semantic chunking and measure whether complaints fall
- ☐ D Run an LLM-as-a-judge model evaluation on the assistant's answers

◆ PAUSE AND DECIDE

What should the team do first?

- A

 Retrieve-only against the ground-truth passages
Answers the exact question asked. Healthy coverage means the retriever is fine and the fault is downstream; poor coverage means the opposite.
- B

 Retrieve-and-generate for an overall score
Confirms quality is poor, which everyone already knows. One end-to-end number cannot say which half caused it.
- C

 Re-ingest with semantic chunking
Commits one of the two available weeks to a fix before establishing that retrieval is the problem. Module 5's ordering, ignored.
- D

 LLM-as-a-judge on the answers
Scores generation quality against context that may already be wrong. A poor answer from poor context looks like a generation failure.

THE SIGNAL

"Ground-truth dataset mapping questions to the passages that should answer them" is the precondition for context coverage. The scenario hands you the one asset retrieve-only requires.

Building a golden dataset

- ◆ **Real questions** from logs, not questions an engineer invented. Users ask things you would not think of.
- ◆ **Stratified.** Cover easy, hard, ambiguous, adversarial and out-of-scope, in roughly production proportions.
- ◆ **Include the ones that should fail.** A question the system must refuse is a test case.
- ◆ **Version it.** Module 10's point: change the set and scores across releases stop being comparable.
- ◆ **Keep it out of any training or prompt data**, or you are measuring memorisation.

ON THE EXAM

A hundred well-chosen stratified questions beat a thousand scraped ones. The value is in coverage of the failure modes you care about, not in raw count.

What running a job actually requires

- ◆ **A JSONL dataset in same-Region S3**, capped at around 1,000 prompts per job.
- ◆ **A service role** trusting Bedrock, with S3 read and write plus model invoke permissions.
- ◆ **Model access for both** the model being evaluated and the evaluator model.
- ◆ **CORS on the output bucket is required only for human-based jobs.** Automatic and RAG jobs do not need it.
- ◆ **Bring your own inference responses** lets you evaluate output from models and pipelines outside Bedrock.

ON THE EXAM

Evaluation is a batch job, not real-time monitoring. It answers 'is this release good enough to ship', while module 15's quality metrics answer 'is production still healthy'.

Turning a score into a pass or fail

- ◆ **Gate on a delta, not an absolute.** "No worse than current" survives a model change.
- ◆ **Establish the noise floor.** Run the same version twice and see how far the score moves.
- ◆ **A threshold inside the noise fails randomly,** and teams re-run until it passes.
- ◆ **Gate on the metric that matters,** not an average. Faithfulness beats tone.
- ◆ **Block on regression, warn on stagnation.** A flat score for six months is a signal.

ON THE EXAM

The noise floor is the step teams skip. Without it you cannot tell a real regression from ordinary variance, and a gate nobody trusts becomes a gate people bypass.

Evaluating an agent is a different problem

- ◆ **Task completion rate.** Did it actually finish the job? The headline agent metric.
- ◆ **Tool usage effectiveness.** Did it pick the right tool, with the right arguments, at the right time?
- ◆ **Reasoning quality in multi-step workflows.** Was the path sensible, or lucky?
- ◆ **Efficiency.** Two agents can both succeed while one takes four steps and the other fourteen.
- ◆ A correct final answer reached by a nonsensical path will fail differently tomorrow. Score the trajectory, not just the outcome.

ON THE EXAM

This is why module 15's iteration count mattered. An agent whose success rate holds while its step count climbs is degrading, and only trajectory-level evaluation sees it.

An operations agent resolves infrastructure tickets using eight tools. Its task completion rate has held steady at 87% for three months and the team considers it healthy.

Costs have risen 60% over the same period with no increase in ticket volume. Investigation shows the average number of tool calls per ticket has climbed from 5 to 13, and several tickets now hit the iteration cap before completing.

What does this indicate, and what should the team evaluate?

- ☐ A Nothing is wrong; completion rate is the metric that matters and it is stable
- ☐ B The agent is degrading. Evaluate trajectories for tool selection accuracy and reasoning quality, and investigate which tool's outputs changed
- ☐ C The model has become slower; switch to a faster model
- ☐ D Raise the iteration cap so more tickets complete

◆ PAUSE AND DECIDE

What does this indicate, and what should the team evaluate?

- A

 Nothing is wrong
Completion rate held while tool calls nearly tripled. The agent is reaching the same outcome by a far worse route, and some tickets no longer finish.
- B

 Degradation. Evaluate trajectories and find the changed tool
Step count is the leading indicator; completion rate is lagging. Something is returning worse information and the agent is compensating by trying harder.
- C

 Faster model
The problem is more calls, not slower calls. A faster model makes thirteen wasteful calls more quickly.
- D

 Raise the iteration cap
Hides the symptom by paying for more of it, and removes the one signal that revealed the problem.

THE SIGNAL

Outcome metrics lag; trajectory metrics lead. A stable success rate on a rising step count is the exact shape of an agent quietly degrading, and only trajectory evaluation explains it.

A/B testing, and the wider ecosystem

- ◆ A/B testing is the only method that measures real user behaviour. Split traffic, compare outcomes, not opinions.
- ◆ Choose a **business outcome** as the metric: escalation rate, edit rate, task completion. Not a judge score.
- ◆ It needs volume and time. A low-traffic assistant will not reach significance on a subtle change.
- ◆ **Shadow deployment** from module 2 evaluates a new version on real traffic with zero user exposure.
- ◆ **RAGAS** and similar open-source frameworks are the bring-your-own alternative when you need metrics Bedrock does not offer.

ON THE EXAM

Bedrock Evaluations covers the common cases well. Reach for an external framework when you need a custom metric, not because managed evaluation feels insufficiently sophisticated.

Reporting results to people who did not run the job

- ◆ **Report the trend, not the number.** A score in isolation is meaningless to a stakeholder.
- ◆ **Tie quality to something the business counts.** Escalation rate, edit rate, resolution time.
- ◆ **Show the failures.** Three real examples of a bad answer communicate more than a percentage.
- ◆ **Amazon Quick** for the stakeholder view, as module 15 established. CloudWatch is for engineers.
- ◆ Never present a judge score as an accuracy figure. It is a comparative measure and will be quoted as fact.

ON THE EXAM

Skill 5.1.8 is explicitly about communicating to stakeholders. The exam asks it because an evaluation nobody acts on has the same value as no evaluation at all.

A team wants continuous quality assurance for a production assistant. It has proposed running a Bedrock evaluation job on every user response in real time, and blocking any response scoring below a threshold before it reaches the user.

Traffic is 60,000 responses a day. The team has budgeted for the evaluator model tokens and believes the approach guarantees quality.

What is wrong with the proposal, and what should be done instead?

- ☐ A Nothing is wrong; this is the correct design for continuous quality assurance
- ☐ B Bedrock evaluation is a batch job and cannot gate a live response. Use guardrail contextual grounding inline, sample responses for asynchronous judging, and run the golden set on a schedule
- ☐ C The cost is too high; reduce the evaluation to 10% of responses in real time
- ☐ D Use a smaller evaluator model to make real-time evaluation affordable

◆ PAUSE AND DECIDE

What is wrong with the proposal, and what should be done instead?

A Nothing is wrong

*Bedrock Evaluations runs as a batch job against a dataset.
It is not an inline API you can place in a request path.*

B Batch, not inline. Guardrails inline, sampled judging, scheduled golden set

Uses the right tool at each layer: contextual grounding is designed to gate a live response, and judging is designed to measure a population.

C Evaluate 10% in real time

*Treats it as a cost problem when it is an architectural one.
The mechanism still cannot gate a response, at any percentage.*

D Smaller evaluator model

Same mistake, cheaper. It also doubles latency on every response, which no scenario would accept.

THE SIGNAL

Module 11's guardrail and module 16's evaluation answer different questions. Contextual grounding gates one response; evaluation measures a release. Neither substitutes for the other.

The five things to carry forward

- ◆ **Measure, do not test.** Many runs, a baseline, and a score that moved more than the noise.
- ◆ **Five job types.** Programmatic, judge, human, retrieve-only, retrieve-and-generate.
- ◆ **Judge from a different model family**, and calibrate against humans. Self-preference bias is real.
- ◆ **Retrieve-only isolates the retriever.** Run it first when quality drops and eliminate half the system.
- ◆ **Score agent trajectories, not just outcomes.** Step count leads; completion rate lags.

ON THE EXAM

Module 17 closes Domain 5 with troubleshooting: the error taxonomy, retrieval failure diagnosis, and a decision tree for where a GenAI system actually broke.